FPGA INT8 MLP ACCELERATOR

Progress Log

Milestones, dated log, open issues, and lessons learned

Team: Joey (`itsKannapi`), Teammate 2 (TBD), Teammate 3 (TBD) · Last updated 2026-10-22 · `https://github.com/itsKannapi/fpga-mlp-accelerator`

How to update this document

Update it in the *same pull request* as the work it describes: flip the milestone status, add a dated entry to the top of the log (newest first), and rebuild the PDF (`latexmk -pdf progress.tex` in `docs/`). Copy the entry template from Section 6. Keep entries factual: what changed, how it was verified, the commit.

1 Team

Member	Focus
Joey (`itsKannapi`)	Project lead; RTL and verification so far
Teammate 2 (TBD)	TBD
Teammate 3 (TBD)	TBD

Names are set once in `docs/preamble.tex` and appear in every document.

2 Milestones

Target: a working, documented on-board demo within four weeks of starting the RTL.

Wk	Milestone	Owner	Status
1	Toolchain: Vivado 2026.1 + license, board smoke test	Joey	DONE
1	Train and quantize the MLP; export weights and golden vectors	Joey	DONE
1	MAC unit + three-tier self-checking testbench	Joey	DONE
2	Layer 1: 32 parallel MACs, ReLU + requantize, bit-exact vs. golden	Joey	DONE
2	Repo made portable (project script, Icarus runner); docs; GitHub	Joey	DONE
2	Layer 2 (32→10) + argmax	Joey	DONE

Wk	Milestone	Owner	Status
2	Full-network testbench: predicted digit + all 10 scores vs. golden	Joey	DONE
3	Image buffer (784 B BRAM) + UART receiver	TBD	TO DO
3	UART transmitter for the prediction; define the host protocol	TBD	TO DO
3	Top-level module + Basys 3 constraints (`.xdc`)	TBD	TO DO
3	Timing closure at 100 MHz	TBD	TO DO
3	Host script: send 100 test images, report on-board accuracy	TBD	TO DO
4	Record Vivado utilization + timing in `design.tex`	TBD	TO DO
4	README polish, block diagram, demo video	TBD	TO DO
–	Binary/ternary (XNOR + popcount) variant reusing the pipeline	TBD	STRETCH

3 Log

Newest first. Commit hashes refer to the `main` branch.

2026-10-22 Full classifier working in simulation

- Added `src/layer2.v` (10 MACs, 34 cycles), `src/argmax.v` (one score per clock, lowest index wins ties, 10 cycles), and `src/mlp_core.v`, which chains everything: **pixels in, digit out, 832 cycles (8.32 µs)**. Streaming with `start` held high: one result every 833 cycles (≈120k images/s).
- Exporter now also writes `rtl/w2_t.mem`; every existing file reproduced byte for byte.
- **Simulated accuracy: 99/100** on the first 100 test images, identical to the Python model (both miss image 8, a 5 read as a 6).
- New testbenches `tb_layer2` (737 checks), `tb_argmax` (4,024), `tb_mlp_core` (2,880). All five testbenches: 14,166 checks, all pass in Icarus. **`tb_mlp_core` also passes in Vivado xsim** (99/100, same latency and throughput).
- 12 planted bugs, all caught. Two findings: a wrong tie-break passes the whole end-to-end test (real scores never tie), so `tb_argmax` is essential; and a stray-start bug was first caught only by luck, so `tb_mlp_core` now watches all stages for 1,000 cycles after each stray start.
- Estimated resources for the full core (Yosys): 42 DSP, 15 RAMB18, 2.2k LUT.
- The project script now works in Vivado 2026.1 (confirmed). Its default tops are now `mlp_core` and `tb_mlp_core`. Repo pushed to GitHub.

- Commit author email fixed to the GitHub `noreply` address and history rewritten once, so commits are credited to the account. All hashes in this log are post-rewrite. Commit `ca57678`.

2026-10-16 Layer 1 verified; repo prepared for the team

- Built `src/layer1.v`: 32 MACs fed by one broadcast pixel per clock and a 256-bit weight word. Latency 786 cycles (7.86 µs at 100 MHz).
- Exporter now also writes `rtl/w1_t.mem` (pixel-major). Re-running it reproduced every existing file byte for byte.
- `sim/tb_layer1.v`: 4,259 checks, **all pass in Vivado xsim**. Mutation testing found that real MNIST images miss three bug classes (border pixels are always 0, no neuron saturates); added synthetic images, and all 10 planted bugs are now caught. Details in `design.tex` §4.
- Estimated resources (Yosys): 32 DSP, 15 RAMB18, ∼1.7k LUT.
- Removed the hard-coded `C:/dev/...` path from the Verilog.
- Added two scripts: `create_project.tcl` rebuilds the Vivado project on any machine, and `run_sims.sh` runs every testbench in Icarus. Started these docs.
- Commits `3782b62` (layer 1) and `6cdeeb6` (docs, portability).

2026-10-09 MAC unit verified

- `src/mac.v`: int8×int8 into a 32-bit accumulator, mapped to one DSP48.
- `sim/tb_mac.v`: 10 directed + 2,000 random + 256 golden checks (2,266 total), **all pass in Vivado xsim** using the real trained weights.
- Three planted bugs each caught; each test tier catches bugs the others miss.
- Commit `d5a2dcb`.

2026-10-08 Board bring-up; model trained and quantized

- Smoke test: a two-switch AND gate on LD0 programmed over JTAG and working. This confirms license, synthesis, implementation, bitstream, drivers.
- Trained the $784 \to 32 \to 10$ MLP: 95.86% test accuracy. int8 version also 95.86% (99.74% of predictions identical). `SHIFT1` = 12.
- Golden vectors for 64 images exported to `rtl/golden.txt`.
- Commit `cf55ba9`.

2026-10-01 Vivado installed

- Vivado 2026.1 needs a license file even for the free tier. Installed the free *Basic Tier, node-locked* license (see `setup.tex`).
- The Artix-7 family was missing from the install; added it through the installer's add-devices mode.

Before RTL **Project scoped**

Chose an int8 MLP accelerator over a RISC-V CPU and over camera/microphone projects: it tells a complete "AI hardware" story and needs no exotic I/O, so it is finishable before internship application deadlines.

4 Open issues

#	Issue	Owner
1	`mlp_fp32.npz` (37 MB, holds the float weights) is not in git, so only Joey's machine can re-run `quantize_export.py`. Fix before anyone needs a new export: commit a small weights-only file and have the script load test data itself.	TBD
2	`tb_layer2` and `tb_argmax` have only run in Icarus. Their modules are covered in xsim through `tb_mlp_core`; run them in xsim when convenient.	TBD

5 Lessons learned

- **"Programmed" in Hardware Manager does not mean your design is on the chip**: it also shows for the factory demo loaded from flash. Use *Program Device*. Each switch lighting its own LED means it's still the factory demo.
- **JTAG programming is volatile**: power-cycling restores the factory demo.
- **Vivado's simulator stops at 1 µs by default**; the project script sets it to run to completion (runtime `-all`).
- **Real data can hide bugs.** MNIST borders are blank, so pixel-skip bugs pass real images. Always add synthetic inputs that touch every path, and use mutation testing to prove the testbench can fail.
- **Some rules need their own test.** Real MNIST scores never tie, so argmax's tie-break can only be checked with made-up inputs.
- **A bug caught by luck is a gap.** If a bug shows up only as a side effect on a later test, make a check that catches it where it happens.
- **Weight layout is part of the interface.** Changing from neuron-major to pixel-major needed a new export file; the testbench checks one layout against the other.

6 Entry template

```
\subsection*{YYYY-MM-DD \quad Short title}
\begin{itemize}
  \item What changed (files, modules).
  \item How it was verified (testbench, checks, xsim or Icarus).
  \item Anything surprising or still open.
  \item Commit \texttt{abc1234}.
\end{itemize}
```